Mayank Gupta

AI Software Engineer

gmayank531@gmail.com · +91-8299828790 · linkedin.com/in/mayank-gupta-4649a6192

Professional Summary

Applied AI and backend engineer with 3+ years shipping production LLM systems: multi-agent orchestration, MCP architecture, RAG / semantic search, and high-throughput Django platforms. Strong in async and distributed processing (AsyncIO, Celery, RabbitMQ), enterprise forecasting, AI workflow automation, and AWS (Lambda, SQS, S3, EC2, RDS, CloudWatch) for serverless, event-driven, monitored production deployments.

Technical Skills

Programming	Python, Django, SQL, Async Programming, AsyncIO, Parallel Processing, Concurrent Processing
AI / GenAI	OpenAI, OpenAI Agents, LangChain, RAG, MCP, Multi-Agent AI Systems, Embeddings, Prompt Engineering, Semantic Search, Vector Databases, Semantic Retrieval Systems, LLM Infrastructure, Applied AI Systems
Backend / Infra	Celery, RabbitMQ, Distributed Systems, Microservices, Event-Driven Systems, Distributed AI Pipelines, Backend Architecture, Scalable AI Infrastructure, Serverless Architecture, Production AI Deployments, AI Workflow Automation
Data / ML	Pandas, NumPy, Scikit-learn, EDA, Data Cleaning, Feature Engineering, Forecasting, Enterprise Forecasting Platforms
Databases	PostgreSQL, MySQL, Amazon RDS, Solr, Vector DBs
Cloud / Tools	AWS Cloud Services, AWS Lambda, Amazon SQS, Amazon EC2, Amazon S3, Amazon CloudWatch, Cloud Monitoring, Linux, Git, VS Code, PyCharm, Jupyter Notebook, Google Colab

Professional Experience

Software Engineer — WNS (Part of Capgemini) — India **Dec 2024 – Present**

- Built enterprise AI analytics, forecasting, and automation on Python/Django: 1800+ dashboards, 16+ AI widgets, and multi-agent / MCP / LLM workflows for KPI analysis, supplier intelligence, summarization, and contextual insights.
- Engineered distributed backends with AsyncIO, Celery, RabbitMQ, semaphores, and 12–18 concurrent workers; processed 50+ dashboard workloads/day via queueing, batching, parallel scheduling, and latency-focused orchestration.
- Integrated AWS Lambda, Amazon SQS, Amazon S3, Amazon EC2, Amazon RDS, and Amazon CloudWatch for serverless execution, messaging, storage, hosting, and monitoring; owned deployment, production support, performance optimization, and environment management.

Associate Software Engineer — Prolifics — India **Jul 2023 – Nov 2024**

- Shipped AI-powered automation and scalable Django backends; built chatbot platforms with RAG pipelines, vector databases, and semantic retrieval.
- Contributed to deployment workflows, debugging, optimization, and production support.

Key Projects

Enterprise AI Analytics & Multi-Agent Orchestration Platform

- Architected a large-scale analytics platform across 1800+ enterprise dashboards and 16+ intelligent AI widgets; designed multi-agent pipelines for KPI analysis, supplier intelligence, forecasting, summarization, and LLM-powered business insight generation.
- Scaled production AI orchestration with AsyncIO, RabbitMQ, Celery, parallel processing, semaphores, and 12–18 workers; AWS Lambda, SQS, S3, RDS, EC2, and CloudWatch for 50+ daily dashboard workloads with concurrency control and efficient task scheduling.

AI-Powered Commodity & Category Forecasting Platform

- Built commodity/category intelligence by integrating news APIs, market signals, and forecasting pipelines; automated ingestion 6 times daily and ran OpenAI GPT prompt pipelines with structured reasoning, web-search enrichment, and dynamic signal weighting.
- Delivered an MCP-style conversational layer with strict commodity/category context isolation; persisted forecast versioning in MySQL (AI overwrites, baseline-to-AI conversion, historical tracking, structured summaries) on Lambda, SQS, S3, RDS, EC2, and CloudWatch.

AI-Powered Audit Automation with OpenAI Agents

- Developed audit automation in Django using OpenAI Agents, LangChain, and RAG: intelligent checklist validation, contextual analysis, implementation-status classification, insight reporting, and transparent workflow documentation for reasoning and audit traceability.

Education

B.Tech – Techno India University, Kolkata CGPA: 8.7

Certifications

Zero to Mastery Python Course – Udemy · Certified Django Course – Udemy · Learn Data Science using ML – Microsoft Ardent · Agentic AI: Build Your First Agentic AI System – LinkedIn Learning